

ME231 Statics Notes

1 Lecture — 24/09/26

Curriculum Position

- Week-1 — Course intro, syllabus, Newton's laws, vectors.

Key Content

1. Mechanics
2. Basic Quantities
3. Idealizations and an intro into rigid body mechanics
4. Newton's Laws of Motion
5. Newton's Law of Gravitational Attraction
6. SI Units
7. Unit Prefixes and Rules for Unit Prefixes
8. Dimensional Homogeneity
9. Sig-Figs and Rounding
10. General Procedure for Analyzing Systems

Instructor Emphasis

- Use 3 Sig-Figs for calculations.
- Take $g = 9.81 \text{ m/s}^2$ for calculations.
- He cares a lot about unit prefix rules.
- It is recommended that calculations be done with the units alongside, so that the resulting unit gets verified.
- Retain a greater number of digits than the problem data for calculations.
- For solving problems with different units, converting to base units first is recommended

Announcements

1. Email appointment before office visit.
2. Lecture notes are provided and said to be adequate.
3. Nonattendance allowed up to 4 days.
4. AA is (100-90) by default,
5. Worked out problems will be provided on the course Teams group.
6. Grading curve is possible based on performance.
7. 1 Midterm (35%), 1 Final (45%), Pop-Quizzes (5% each) Announced a week before.
8. Textbook: Hibbeler, R.C. Engineering Mechanics
9. Next week will consist of 3 sessions.

Open Questions

Independent Follow-up

- Minimum: Have a light read.
- Exploration: Watch some stuff on statics and FEA